

Monte Carlo Integration Simulation

Objective:

Estimate the integral of $\sin(x)$ from 0 to π using Monte Carlo simulation.

Method:

- Generate 10000 random points in rectangle $[0, \pi] \times [0, 1]$
- Count points under the curve
- Estimate integral = area * (points under curve / total points)

Result:

Estimated integral = 1.977004

